

4th LIMIT Workshop

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Keywords: {Self, Semi, Weakly, Un}-supervised learning and {Zero, One, Few}-shot learning with very limited resources, Training with synthetic images

1 Workshop title

Representation Learning with Very Limited Resources: When Data, Modalities, Labels, and Computing Resources are Scarce (LIMIT)

2 Relevance of topic

Abstract. The LIMIT Workshop takes the position that, while scaling laws have driven dramatic progress in computer vision, representation learning under limited resources provides a crucial opportunity to revisit its fundamental principles. This workshop explores how foundation models can be learned and evaluated under limited labels, data, parameters, and computational resources. In recent years, computer vision has largely evolved under the “*bigger is better*” paradigm, characterized by increasing data, parameters, and computational scale. However, this scaling assumption does not universally hold across domains and industrial settings. As computer vision expands beyond vision-language models toward spatial understanding and robotic manipulation, new models such as VGGT and π_0 are emerging. However, the high-quality data they require remains limited. Moreover, large-scale foundation models continue to face challenges related to data provenance, licensing, and bias, which expose unresolved issues inherent in real-world datasets. LIMIT Workshop seeks to foster a collaborative community spanning academia and industry and to promote democratic and reproducible research on foundation models, even when large-scale training is not feasible.

Topics:

- Building vision/spatial multi-modal models under limited-resources settings
- Synthetic training strategies for vision/spatial multi-modal models
- Issues (social bias, privacy and transparency) with real data and labels
- Vision+X and multi-modality to accelerate a learning efficiency
- Analysis for vision/spatial multi-modal models under the LIMIT philosophy

Primary contact name and email	Ryousuke Yamada / ryosuke.yamada@aist.go.jp
Half or full day	Half day
Anticipated audience size	150
Requested number of poster boards	20
Papers published in proceedings?	Yes
Special requests	No, special requests at the submission

3 Relevance of organizers

- Ryousuke Yamada and Hirokatsu Kataoka launched LIMIT at ICCV’23 and have since organized consecutive editions (ICCV’23, CVPR’24, ICCV’25). We also contributed LIMIT-related research, including formula-driven supervised learning (ACCV’20 and BMVC’23) and built LIMIT.lab¹ as a hub for learning under limited resources, connecting researchers.
- The team has extensive workshop experience in the computer vision community, including FOUND (Yoshihiro; ICCV’25), multiple workshops (Yuki; ICCV’23/ECCV’20/24), and MDALC (Nakamasa; ICCV’19), demonstrating our ability to recruit speakers, manage reviews, and run the event smoothly.
- Organizers contribute to key ECCV areas is that efficient learning, generative modeling, and 3D vision: Junhwa, Shumpei, and Daichi (generative models, 3D scene understanding, zero-shot depth estimation), and Naoshi and Jona are working on developing and analyzing of vision foundation models. Moreover, Su has been researching on efficient learning for vision models. This breadth helps curate a timely, high-quality program aligned with LIMIT.

4 Tentative program outline

Format: in-person

Expected workshop size: Middle (150 attendees)

Request for full-day workshop: No

Schedule: The workshop will contain 4 invited talks, 1 oral sessions, and 1 poster session (expected 20 posters) as shown in Table 1. We are able to organize the workshop on either date during the ECCV 2026 workshop days. We will flexibly arrange the time schedule along with the invited speakers.

Paper submission: We invite submissions of original research papers related to the workshop topics. The review process will involve over 100 reviewers from previous LIMIT workshops and the broader LIMIT.Lab community.

Paper review timeline (tentative):

- Submission deadline: July 11th, 2026
- Review deadline: July 25th, 2026
- Notification: August 5th, 2026

The workshop will accept papers of up to 8 pages for publication in the Springer proceedings. All submissions will be peer-reviewed in a double-blind manner and must be formatted according to the ECCV 2026 guidelines.

Competition: The workshop does not plan to associate a competition.

¹ limitlab.xyz

Table 1. Workshop Schedule

Time	Duration	Event
9:00 - 9:10	10 min	Opening remarks
9:10 - 9:40	30 min	Invited talk (1)
9:40 - 10:10	30 min	Invited talk (2)
10:10 - 10:20	10 min	Short break
10:20 - 10:50	30 min	Invited talk (3)
10:50 - 11:20	30 min	Invited talk (4)
11:20 - 12:00	40 min	Oral session
12:00 - 12:10	10 min	Closing remarks
12:10 - 13:30	80 min	Lunch & Poster session (20 posters)

4.1 List of invited speakers

- **Björn Ommer** (Ludwig Maximilian University of Munich)
 - Confirmed
 - Homepage link: <https://ommer-lab.com/ommer/>
 - Relevance to the workshop: He conducts research on efficient representation learning and generative modeling in compressed and latent spaces, enabling high-quality synthesis under limited computational budgets (*e.g.*, DisMo: Disentangled Motion Representations for Open-World Motion Transfer (NeurIPS 2025), TREAD: Token Routing for Efficient Architecture-agnostic Diffusion Training (ICCV 2025), Does VLM Classification Benefit from LLM Description Semantics? (AAAI 2025)).
- **Ranjay Krishna** (University of Washington / Allen Institute for AI)
 - Confirmed
 - Homepage link: <https://www.ranjaykrishna.com/>
 - Relevance to the workshop: His research focuses on efficient multimodal learning for vision-language models and embodied AI (*e.g.*, MolmoAct: Action Reasoning Models that can Reason in Space (ICRA 2026), Scale Can’t Overcome Pragmatics: The Impact of Reporting Bias on Vision-Language Reasoning (TACL 2026), and Molmo and PixMo: Open Weights and Open Data for State-of-the-Art Multimodal Models (Best Paper Honorable Mention at CVPR 2025)).
- **Jia Deng** (Princeton University)
 - Confirmed
 - Homepage link: <https://www.cs.princeton.edu/~jiadeng/>
 - Relevance to the workshop: He has conducted research on data-efficient visual recognition and representation learning, including scalable dataset construction and synthetic data pre-training (*e.g.*, Zero-shot Sim2Real Transfer for Magnet-Based Tactile Sensor on Insertion Tasks (ICRA 2026), Seeing and Seeing Through the Glass: Real and Synthetic Data for Multi-Layer Depth Estimation (ICCV 2025), and Infinigen Indoors: Photorealistic Indoor Scenes using Procedural Generation (CVPR 2024)).

– **Anna Kukleva** (Meta)

- Confirmed
- Homepage link: <https://annusha.github.io/>
- Relevance to the workshop: Her research centers on representation learning for images and videos under limited supervision, spanning self-supervised learning, synthetic data generation, and open-set recognition (*e.g.*, VideoGEM: Training-free Action Grounding in Videos (CVPR 2025), T-FAKE: Synthesizing Thermal Images for Facial Landmarking (CVPR 2025), and OrCo: Towards Better Generalization via Orthogonality and Contrast for Few-Shot Class-Incremental Learning (CVPR 2024)).

5 Diversity

We have considered personal characteristics such as nationality, affiliations, and positions. Specifically, the organizers and the invited speakers come from about 10 countries/regions, 8 universities, 3 companies, and 3 national research institutes. Also, some students have experience running hybrid (physical and virtual) conferences of 1,000 attendees. We have accepted 4 invited speakers in the LIMIT workshops and considered diverse keynote topic coverage. The speakers cover varying backgrounds, including generative models, vision/language foundation models, 3D vision, syntetic training, self-supervised learning.

6 Broader impacts

6.1 Social considerations based on the organizer’s research

Recent progress in computer vision has been largely driven by scaling laws: increasing dataset size, model capacity, and computational resources. While this paradigm has led to remarkable performance gains of foundation models, it has also exposed significant ethical and societal concerns, including data bias, privacy violations, transparency associated with large-scale training, and unequal access to computational infrastructure. LIMIT Workshop explores research directions that challenge the assumption that progress must rely on ever-increasing scale. By promoting learning under limited labels, data, parameters, and computational resources, the workshop aims to encourage more sustainable, accessible, and responsible AI development. In particular, emerging approaches such as self-supervised learning, formula-driven supervision, and synthetic training suggest that competitive visual representations can be learned with substantially reduced reliance on massive real-world datasets. These directions hold promise for mitigating ethical risks tied to data collection while broadening participation in AI research. We view advancing research on learning under limited resources not only as a technical challenge but also as a societal necessity. Through invited talks and paper presentations, we aim to foster critical and constructive dialogue on how computer vision research can progress without reinforcing existing inequities in access to data and computational resources.

6.2 Ethical Considerations

Throughout the past our workshop organization, we have formed a LIMIT.Lab in order to support, e.g., researchers and engineers in AI construction with limited resources and gather them in one place. The organizers and the LIMIT.Lab actively engage in discussions on foundation models under limited resources, particularly in relation to large-scale dataset collection, data transparency, and the environmental and societal implications of scaling foundation models. Research under limited-resource settings may contribute to mitigating certain ethical risks associated with large-scale training. For example, learning from smaller, curated datasets can improve transparency and facilitate the removal of problematic or unethical data. Similarly, synthetic or simulated pre-training may reduce reliance on web-scale data collection, thereby lowering risks related to copyright, privacy, and bias. Moreover, exploring data-efficient and compute-efficient learning strategies can help democratize foundation model research by lowering hardware and infrastructure barriers. While limited-resource approaches do not replace large-scale training paradigms, they offer complementary pathways toward more sustainable, transparent, and accessible AI development.

7 Related previous workshops

- **Efficient learning.** Efficient Deep Learning for Computer Vision (CVPR 2025/2026), Efficient Large Vision Models (CVPR 2024/2025), Curated Data for Efficient Learning (ICCV 2025) → In addition to efficient deep learning, we welcome papers with a broader scope, that extends to data, labels, model parameters, and computational efficiency as constraints.
- **Synthetic data.** EMACS (CVPR 2025), SyntaGen (CVPR 2024/2025), Synthetic Data for Computer Vision Workshop (CVPR 2025), SynData4CV (CVPR 2024), Will Synthetic Data Finally Solve the Data Access Problem? (ICLR2025) → We consider, but are not limited to, the topic of synthetic data. While training with synthetic data is an area of interest, we will also encourage broader training strategies with a focus on limited resources.
- **AI ethics.** Workshop on Responsible Generative AI (CVPR 2025), Fair, Data-efficient, and Trusted Computer Vision (CVPR 2025), FAILED (ICCV 2025), ABAW (ICCV 2025), Privacy for Vision & Imaging (ECCV 2024) → We deal with a learning framework that fundamentally solves dataset-related problems such as fairness, accountability, ethics, transparency, and privacy.
- **Un-/self-supervised learning.** SSL-WIN (ECCV 2022/2024), Self-supervised Learning – Theory and Practice (NeurIPS 2024), → We will also discuss the fundamental topic in order to solve dataset-related issues. Yuki M. Asano has been the primary organizer in both workshops.
- **Limited labels.** L3D-IVU (CVPR 2024), L2ID (ECCV 2022), and Self-Improving Foundation Models Without Human Supervision (ICLR 2026) → Our workshop addresses not only the issue of supervision but also the limited data. We also deal with an extreme setting (single image/video) and formula-driven approaches.